

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0433

Wood Acres 138 kV
March 24, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0433. The Interconnection Customer proposes a 304.0 MW solar photovoltaic facility interconnecting at the Wood Acres 138 kV substation in the Maryland area. The request was received on February 12, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 252 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,534,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BEVERLY FARMS - BELLS MILL 138.0 kV	TABER - GLENANNAN 345.0 kV	140.4	212	7.3
BEVERLY FARMS - PHILLIPS PUMP 1 138.0 kV	TABER - GLENANNAN 345.0 kV	135.5	250	5.1
DOUCETTE - WOOD ACRES SUBSTATION 138.0 kV	WOOD ACRES - SARANAC FACILITY 138.0 kV Line	135.2	192	100.0
JEROME - FAIRLAND 138.0 kV	FAIRLAND - SIUC 138.0 kV line	133.9	277	5.0
WOOD ACRES - SARANAC FACILITY 138.0 kV	DOUCETTE - WOOD ACRES 138.0 kV L	132.5	196	100.0
DARIEN - SLIGO SUBSTATION 138.0 kV L	TABER - GLENANNAN SUBSTATION 345.0 kV line	131.0	366	8.1
DARIEN - COLESVILLE 138.0 kV	GLENANNAN SUBSTATION - TABER 345.0 kV	128.1	487	6.6
SLIGO - LINDEN 138.0 kV line	TABER - GLENANNAN 345.0 kV Line	127.4	398	8.1
DRUM - MONTPELIER 138.0 kV	DIXLETA - TABER 345.0 kV	127.0	432	6.6
DIXLETA - N POLE 138.0 kV Circuit 1	DIXLETA - NORTH POLE 138.0 kV Cir. 2	125.7	465	7.4
DIXLETA - NORTH POLE 138.0 kV Circuit 2	N POLE - DIXLETA 138.0 kV	123.6	472	7.4
HUGHSON - DIXLETA 345.0 kV	RUSTIC - DIXLETA 345.0 kV	120.8	1,673	22.6
JEROME - HIGH RDGE SUBSTATION 138.0 kV L	TABER - GLENANNAN 345.0 kV L	120.3	830	10.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HIGH RIDGE - NORT POLE 138.0 kV	GLENANNAN - TABER 345.0 kV L	120.2	799	10.3
TOKOMA - YULEE 138.0 kV	COLESVILLE - DARIEN 138.0 kV Line	118.7	243	14.3
FAIRLAND - SUNBRIGHT 138.0 kV Line	GLENANNAN - TABER 345.0 kV	118.5	858	11.2
RUSTIC - DIXLETA 345.0 kV	HUGHSON - DIXLETA 345.0 kV Line	118.0	1,612	20.5
RUSH - COLESVILLE 138.0 kV	DARIEN - COLESILLE 138.0 kV	117.0	374	5.0
TABER - GENANNAN 345.0 kV	SIERRA PACIFIC LUMBER 138/1 kV Autotransformer	116.4	2,247	21.4
SUNBRIGHT - COLESVILLE 138.0 kV	TABER - GLENANNAN 345.0 kV	116.4	922	11.2
WESTPORT - WHEELABRATOR BALTIMORE REFUSE 138.0 kV	TABER - DIXLETA 345.0 kV	116.1	589	6.4
TABER - DIXLETA 345.0 kV	SIERRA PACIFIC LUMBER 138/1kV XFMR	113.7	2,150	19.6
DIXLETA - DRUM 138.0 kV	GLENANNAN - TABER SUBSTATION 345.0 kV	113.1	799	6.7
SIUC - FAIRLAND 138.0 kV	TABER - GLENANNAN 345.0 kV	112.5	643	7.3
LAKE CRK SUBSTATION - SIERRA PACIFIC LUMBER 138.0 kV L	TABER - GLENANNAN 345.0 kV	111.0	782	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEVERLY FARMS - BELLS MILL 138.0 kV	Rebuild line with higher-capacity conductor	\$1,802,000	18% (\$323,000)
PHILLIPS PUMP 1 - BEVERLY FARMS 138.0 kV	Rebuild line with higher-capacity conductor	\$4,769,000	26% (\$1,246,000)
DOUCETTE - WOOD ACRES SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$2,857,000	100% (\$2,857,000)
JEROME - FAIRLAND 138.0 kV	Rebuild line with higher-capacity conductor	\$8,452,000	28% (\$2,405,000)
WOOD ACRES - SARANAC FACILITY 138.0 kV	Rebuild line with higher-capacity conductor	\$2,571,000	100% (\$2,571,000)
DARIEN - SLIGO 138.0 kV	Rebuild line with higher-capacity conductor	\$4,256,000	25% (\$1,056,000)
DARIEN - COLESVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$944,000	9% (\$86,000)
LINDEN - SLIGO 138.0 kV	Rebuild line with higher-capacity conductor	\$1,041,000	25% (\$258,000)
DRUM - MONTPELIER 138.0 kV	Rebuild line with higher-capacity conductor	\$8,778,000	4% (\$312,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DIXLETA - N POLE 138.0 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,504,000	3% (\$52,000)
DIXLETA - N POLE 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,782,000	3% (\$61,000)
HUGHSON - DIXLETA 345.0 kV	Rebuild line with higher-capacity conductor	\$43,065,000	2% (\$1,003,000)
JEROME - HIGH RDGE SUBSTATION 138.0 kV L	Rebuild line with higher-capacity conductor	\$2,453,000	3% (\$66,000)
HIGH RIDGE - NORT POLE 138.0 kV	Rebuild line with higher-capacity conductor	\$2,847,000	3% (\$76,000)
TOKOMA - YULEE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,399,000	8% (\$109,000)
SUNBRIGHT - FAIRLAND 138.0 kV	Rebuild line with higher-capacity conductor	\$478,000	3% (\$16,000)
RUSTIC - DIXLETA 345.0 kV	Rebuild line with higher-capacity conductor	\$46,871,000	2% (\$1,081,000)
RUSH - COLESVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$4,565,000	7% (\$334,000)
TABER - GENANNAN 345.0 kV	Rebuild line with higher-capacity conductor	\$38,020,000	2% (\$848,000)
SUNBRIGHT - COLESVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,511,000	3% (\$50,000)
WESTPORT - WHEELABRATOR BALTIMORE REFUSE 138.0 kV Line	Rebuild line with higher-capacity conductor	\$813,000	3% (\$22,000)
TABER - DIXLETA 345.0 kV	Rebuild line with higher-capacity conductor	\$18,992,000	2% (\$419,000)
DIXLETA - DRUM 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,610,000	2% (\$39,000)
SIUC - FAIRLAND 138.0 kV	Rebuild line with higher-capacity conductor	\$1,866,000	7% (\$126,000)
LAKE CREEK - SIERRA PACIFIC LUMBER 138.0 kV	Rebuild line with higher-capacity conductor	\$3,364,000	4% (\$118,000)
	Total allocated to project		\$15,534,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.